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SEMESTER - 6TH

DEPERMENT - EE

SUBJECT CODE - PC EE-601

SUBJECT - POWER SYSTEM II

1. Write a report on Relay operation, relay types, relay structure

REPORT ON RELAY OPERATION, RELAY TYPES AND RELAY STRUCTURE

Introduction :- A relay is an electrically operated switching device used to control a high power circuit using a low-power signal. Relays are widely used in electrical power systems for protection, control, and automation purposes. They help in detecting faults and isolating faulty sections to ensure safe and reliable operation of electrical equipment.

Relay operation :- The basic working principle of a relay is based on electromagnetic attraction.

When the current flows through the relay coil - A magnetic field is produced. The magnetic field attracts the armature. The armature movement changes the position of the contacts (NO/NC). The circuit is either completed or broken.

When the coil is de-energized :- The magnetic field collapses. A spring pulls the armature back to its original position and the contacts return to their normal state.

Relay Structure :- A typical electromagnetic relay consists of the following parts :-

1. Coil (Electromagnet) - Produces magnetic field when energized.
2. Iron Core - Concentrates and strengthens the magnetic field.
3. Armature - Movable iron piece attracted by the magnetic field.

4. Spring - Return the armature to normal position when coil is de-energized.

5. Frame - protects internal parts from damage and dust.

In modern relays, especially digital relays, electronic circuit, microprocessors, and communication ports are also included.

TYPES OF Relays :- Relays are classified based on operation and applications.

A) Based on Construction -

1. Electromagnetic Relay
2. Static Relay
3. Digital Relay
4. Numerical Relay

B) Based on function -

1. Overcurrent Relay
2. Overvoltage Relay
3. Undervoltage Relay
4. Differential Relay
5. Distance Relay
6. Directional Relay

CONCLUSION :- Relays play a vital role in electrical power systems by detecting abnormal conditions and protecting equipment from damage.

3. Two generators rated 10 MVA, 13.2 kV and 15 MVA, 13.2 kV are connected in parallel to a bus bar. They feed supply to 2 motor of inputs 8 MVA and 12 MVA respectively. The operating voltage of motors is 12.5 kV. Assuming the base quantities as 50 MVA, 13.8 kV, draw the per unit reactance diagram. The percentage reactance for generators is 15% and that for motors is 20%.

→ Given Data —

Generators:-

$$G_1 = 10 \text{ MVA}, 13.2 \text{ kV}, X = 15\%$$

$$G_2 = 15 \text{ MVA}, 13.2 \text{ kV}, X = 15\%$$

Motors:-

$$M_1 = 8 \text{ MVA}, 12.5 \text{ kV}, X = 20\%$$

$$M_2 = 12 \text{ MVA}, 12.5 \text{ kV}, X = 20\%$$

Common Base:-

$$\text{Base MVA} = 50 \text{ MVA}$$

$$\text{Base voltage} = 13.8 \text{ kV}$$

Convert percentage Reactance to per unit —

$$X_{pu} = \frac{\%X}{100}$$

$$X_G = \frac{15}{100} = 0.15 \text{ pu}, \quad X_M = \frac{20}{100} = 0.20 \text{ pu}$$

Convert to New Base —

$$X_{pu} = X_{pu} \times \frac{\text{MVA}}{\text{MVA}} \times \left(\frac{KV_{old}}{KV_{New}} \right)^2$$

Generator 1 (10 MVA, 13.2 kV)

$$X_{G1} = 0.15 \times \frac{50}{10} \times \left(\frac{13.2}{13.8} \right)^2$$

$$= 0.5 \times 5 \times 0.014$$

$$= 0.686 \text{ pu}$$

Generator 2 (15 MVA, 15%); -

$$X_{G2} = 0.15 \times \frac{50}{15} \times (0.956)^2$$

$$= 0.15 \times 3.33 \times 0.914$$

$$= 0.457 \text{ pu}$$

Motor 1 (8 MVA, 12.5 kV); -

$$X_{M1} = 0.20 \times \frac{50}{8} \times \left(\frac{12.5}{13.8}\right)^2$$

$$= 0.20 \times 6.25 \times 0.821$$

$$= 1.026 \text{ pu}$$

Motor 2 (12 MVA, 12.5 kV); -

$$X_{M2} = 0.20 \times \frac{50}{12} \times 0.821$$

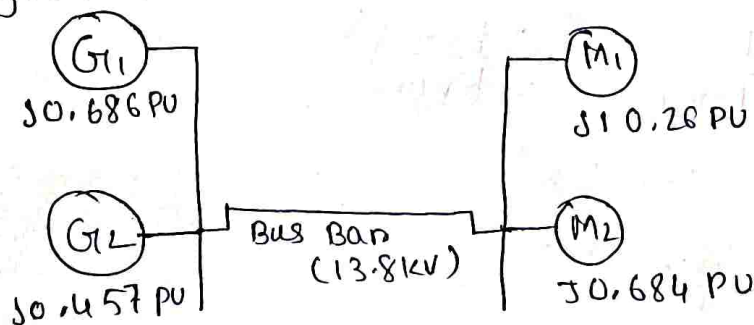
$$= 0.20 \times 4.167 \times 0.821$$

$$= 0.684 \text{ pu}$$

Final per unit Reactances :-

Equipment	Reactance
G1	0.686 pu
G2	0.457 pu
G3	1.026 pu
G4	0.684 pu

Diagram :-



4. For the following Power System network shown in the figure, the Specification of the Component are as follows:-

G_1 :- 25 KV, 100 MVA, $X = 9\%$

G_2 :- 25 KV, 100 MVA, $X = 9\%$

T_1 :- 25/220 KV, 90 MVA, $X = 12\%$

T_2 :- 220/25 KV, 90 MVA, $X = 12\%$

Transmission line 1: 220 KV, $X = 150 \text{ Ohms}$. Draw the Reactance diagram.

→ generators are rated 100 MVA, Choose,

Base MVA = 100 MVA

Base voltage at generator = 25 KV

Base voltage at Transmission = 220 KV

transformer ratio = $\frac{25}{220} \text{ KV}$

Convert all reactance to 100 MVA Base -

Generators (G_1 and G_2) :-

given, 100 MVA, 25 KV
 $X = 9\%$

$$X_G = \frac{9}{100} = 0.09 \text{ pu}$$

Base MVA = rated MVA - No Conversion needed.

$$\therefore X_{G1} = X_{G2} = 0.09$$

Transformers (T_1 and T_2) :-

Given, 90 MVA, $X = 12\%$.

$$X_{\text{new}} = X_{\text{old}} \times \frac{\text{MVA}_{\text{base}}}{\text{MVA}_{\text{rated}}}$$

$$X^T = 0.12 \times \frac{100}{90} = 0.133 \text{ pu}$$

So,

$$X_{T1} = X_{T2} = 50.133$$

Transmission line:-

Given, voltage = 220 kV

Reactance = 150 Ω

first Calculate base Impedance:-

$$Z_{base} = \frac{(KV_{base})^2}{MVA_{avg}} = \frac{(220)^2}{100} = \frac{48400}{100}$$

$$Z_{base} = 484 \Omega$$

Now Convert to per unit ; $X_{line} = \frac{150}{484} = 0.31 \text{ pu}$

Final Per unit Reactance:-

Component	Reactance
G_1	50.09
G_2	50.09
T_1	50.133
line	50.31
T_2	50.133

Diagram:-

